Rigid Motion

Let V be an inner product space over $\mathbb{R}$.

Def. A map $f: V \to V$ is called a Rigid Motion if: for all $x, y \in V$,

$$\|f(x) - f(y)\| = \|x - y\|$$

Thm. Let $f: V \to V$ be a Rigid Motion. Then,

there exists unique orthogonal operator $T: V \to V$ and translation $g: V \to V$ s.t. $f = g \circ T$.

Proof. Define

$$T: V \to V : x \mapsto f(x) - f(0) \quad \text{and} \quad g: V \to V : y \mapsto y + f(0)$$

Then, $g \circ T(x) = g(T(x)) = g(f(x) - f(0)) = f(x) - f(0) + f(0) = f(x)$. $\therefore g \circ T = f$.

1). T preserves a Norm.

Let $x \in V$ be given.

$$\|T(x)\| = \|f(x) - f(0)\| = \|x - 0\| = \|x\|.$$

2). T preserves inner product.

Let $x, y \in V$ be given.

$$\|x - y\|^2 \underset{T \text{ is Rigid}}{=} \|T(x) - T(y)\|^2 = \|T(x)\|^2 + \|T(y)\|^2 - 2\langle T(x), T(y)\rangle$$
$$= \|x\|^2 + \|y\|^2 - 2\langle T(x), T(y)\rangle$$
$$= \|x\|^2 + \|y\|^2 - 2\langle x, y\rangle$$

Thus $\langle T(x), T(y)\rangle = \langle x, y\rangle$

3). T is linear.

Let $x, y \in V$ and $\alpha \in \mathbb{R}$ be given.

$$\|T(\alpha x + y) - (\alpha T(x) + T(y))\|^2 = \|T(\alpha x + y) - T(y)\|^2 + \|\alpha T(x)\|^2 - 2\langle T(\alpha x + y) - T(y), \alpha T(x)\rangle$$
$$= \|\alpha x + y - y\|^2 + \alpha^2\|x\|^2 - 2\alpha(\langle T(\alpha x + y), T(x)\rangle - \langle T(x), T(y)\rangle)$$
$$= 2\alpha^2\|x\|^2 - 2\alpha(\langle \alpha x + y, x\rangle - \langle x, y\rangle)$$
$$= 2\alpha^2\|x\|^2 - 2\alpha^2\|x\|^2 = 0.$$

4). g, T unique.

If $f = T + a = U + b$ for some linear T, U and Scalar a, b,

then $f(0) = a = b$. Thus $T = U$.